Automatic Scripts Retrieval and Its Possibilities for Social Sciences Support Applications

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Abstract. This paper introduces our method for automatic Schankian-like scripts retrieval from the Internet resources and its preliminary results which might be interesting for Social Sciences researchers. We describe the first module of our system, which is supposed to automatically retrieve commonsensical knowledge from the Web resources by using web-mining techniques. It retrieves minimal "object — action — action" scripts which show humans' common activities changing due the origin of a webpage author. Such data can be used in fields of economics, psycholinguistics, sociolinguistics, psychology, sociology or in language education. By this paper we would like to make NLP researchers notice the potential of commonsense retrieval and encourage them to consider creating such tools for their languages.
Keywords: commonsense, web-mining, social sciences support.

1 Introduction

1.1 Need for Commonsense Retrieval

Nowadays, amount of information increases quickly with rapid growth of the Internet. As an enormous database, it is convenient if the information which humans need is acquired from the huge source of World Wide Web. Related information can be retrieved in a matter of seconds but it is mostly knowledge which we, humans, need. Obvious knowledge called "commonsense" does not become an object of search queries as we do not need it. However the computers are told to lack this knowledge which we gather our whole lives and it is a reason that people do not treat machines as intelligent partners, especially during conversations. There are several researches of gathering commonsense as CyC [1] or OpenMind Commonsense [2]. CyC contains over a million hand-crafted assertions [3] and OpenMind commonsense enabled construction of a 700,000 assertion commonsense knowledge base, gathered through a web community of collaborators. But they concentrate on manual or half-automatic processing and are done only for English language. As we assume that there is too much of such knowledge to be inputted, we try to

make this process automatic by using Web-mining techniques[1]. As we claim that Japanese language to have the best predispositions for such processing thanks to its particles, we concentrate on Japanese WWW resources. However

Table 1. Main Japanese particles and their functions

Particle	Role
WA	Topic-Indicating
GA	Linking-Indicating
NO	Possessive-Indicating
WO	Object-Indicating
NI	Direction-Indicating
DE	Place or Means of Action-Indicating
HE	Destination-Indicating
TO	Connective
MO	Addition-Indicating
YORI	Comparison-Indicating
NODE	Reason-Indicating
V-KARA	Reason Indicating
N-KARA	Lower Limit-Indicating
MADE	Upper Limit-Indicating
DEMO	Emphasis-Indicating

such research does not have to be restricted to Japanese, based on the same principles, an application could work with other languages – for example by using prepositions in English or regular expressions for non-gender counting in Polish. Before we started our commonsense retrieval project, we wanted to observe how is Japanese usable for our purposes and how the commonsense differs depending on a language. In this paper, we introduce our definition of a script, our algorithm, the basic system architecture, the experiment and the results. Then we compare Japanese results with Chinese, Polish and English. This paper introduces our preliminary system with small sets of data but we decided it is enough to introduce the results to the NLP colleagues from different language areas.

1.2 Definition of a Script

For retrieving scripts from WWW we had to simplify the original script [4] definition. Here, the scripts are the memory structures which summarize a typical phenomenon sequence into one frame, and it is defined as what expressed the standardized action which human is performing every day in the form of the chain of a phenomenon in alignment with the passage of time. For

[1] by which we mean here text-mining techniques using data sets being retrieved from WWW instantly.

example, the "restaurant script" can usually express that the customer "enters into restaurant", "sits down on a chair", "places an order", "eats some food", "pays money" and "leaves". As it is the very beginning of our research, we treat any verb chain as a script if it has at least one noun (object) with a particle joining a noun and a verb. Therefore the smallest unit as "watch the TV then go to bed" we also treat as a script and first we experimented with such simple units calling them "minimal scripts".

2 Our System

2.1 Algorithm

In the beginning of our research, we decided to work with nouns as keywords for collecting the scripts. We retrieved them from sentences and extracted the related sequences for creating dictionaries as verb dictionary, noun dictionary and n-gram dictionaries using 1,907,086 sentences WWW corpus retrieved by Larbin robot. The noun and verb dictionaries consist of 79,460 verbs and 134,189 nouns retrieved with help of morphological analyze tool for Japanese — ChaSen [5]. For creating scripts automatically, our system must search for the relationships between verbs and nouns and also between particular verbs. In this step, we used the verbs and nouns which had the highest occurrence, as they are often used in our everyday lives, for example *television*, *movie*, *food*. As mentioned above, we used Japanese language, which has useful grammar features like *rentaikei* where the verb suffix *te* usually joins verbs in a time sequence e.g. *gohan wo tabe*-**te** *neru* (to sleep after having a dinner). We can distinguish and change verb suffixes of Japanese to retrieve other, for example casual, strings for the commonsense, however here we concentrated on *rentaikei* for collecting actions following one after another. This process is the first step for the more complex categorization–using scripts creation proposed by Rzepka at al. [6]:

(dinner, chicken soup, apple...)–*wo* [2]-eat–*te*→
then
(cafeteria, restaurant, garden...)–*wo*–leave–*te*→...

Objects and actions (nouns and verbs joined by a particle) can easily create categories for objects. For example, "to eat" creates a food category, and "to leave" creates a place category. We also can use the pair of verbs to make a category smaller or combine to make a new script – "sit" and "fly" gather other objects than "sit" and "eat".

2.2 Architecture

Basically, our system's architecture for creating scripts can be summarized into the following processing steps:

[2] *wo*: object-indicating particle, see Table. 1.

a) The user inputs a noun as a keyword;
b) The system uses the web corpus for frequency check to retrieve 3 most frequent verbs following the keyword noun;
c) The most frequent particle between noun keyword and 3 most frequent verbs is discovered;
d) For creating bi-gram the system retrieves a list of most frequent verbs occurring after the previously chosen verb;
e) By using Google resources [7], the system checks if the noun-particle unit occurs with new verb-verb unit;
f) If yes — the minimal script is remembered in frequency order, as following:

$$\mathbf{Ms} = \mathbf{N} + \mathbf{P}_{\max} + \mathbf{V}_{\max 1} + \mathbf{V}_{\max 2}$$

N: *Noun keyword;*
$P_{\max}$: *the most frequent particle joining noun and verb;*
$V_{\max 1}$: *most frequent verb occurring after the* N*;*
$V_{\max 2}$: *most frequent verb occurring after* $V_{\max 1}$*;*

2.3 Experiment and Results

The steps a)-c) showed that text processing on our Web corpus takes too much time (average 52 seconds for a query) and for longer sequences (steps d and e) even bigger version of Web corpus (9,802,434 sentences) did not have enough data, therefore we decided to use Google API. Because of problems with Unicode we gave up Linux and made next steps on Windows system (Pentium 4, CPU 3GHz, 512 MB RAM) using Perl for the text processing and Java for connecting to Google servers. For example, retrieving an action following "watching TV" took 3 seconds (165 hits) but selecting the first pair of noun and verb still must be done with our corpus due to the Google limitations (1,000 searches per day) and takes up to one hour in very common keywords like *book*. We considered three most frequent "following actions" as commonsensical and inputted only common nouns (*subway, book, work, newspaper, baseball, bath, flu, swimming, water, bread, movie, television, music, home, car, word* and *meal*). We managed to confirm the naturalness of all retrieved trigrams based on them, though it was not always possible to collect the whole set of three verbs due to the Google limits and encoding errors, as in a "baseball" case in the examples below:
— fill with **water** and (boil, use, mix)
— get on the **subway** and (go back home, move, go)
— eating **meal** and (go back home, sleep, lose weight)
— play **baseball** (make living)
— take a **bath** and (*encoding error*, sleep, go out)
— hear a **music** and (feel happy, cry, to be touched)

— read a **book** and (spend time, learn, think)
— go back **home** and (sleep, see, have a meal)
— drive a **car** and (go back home, move, run away)
Here is a specific example of a case where "TV" was the keyword:
— terebi *wo* mite neru — {*watching TV* → *sleeping*}: 165 times
— terebi *wo* mite warau — {*watching TV* → *laughing*}: 90 times
— terebi *wo* mite omou — {*watching TV* → *thinking*}: 74 times
Because the three above-mentioned actions were most frequent, we assumed that they are the minimal scripts. During using the Google API, we discovered that the data set for Google API apparently differs from the data set used by Google search engine but we confirmed that the percentage of occurrences is similar. The difference rate of Google API and Google search engine is shown in Fig.1.

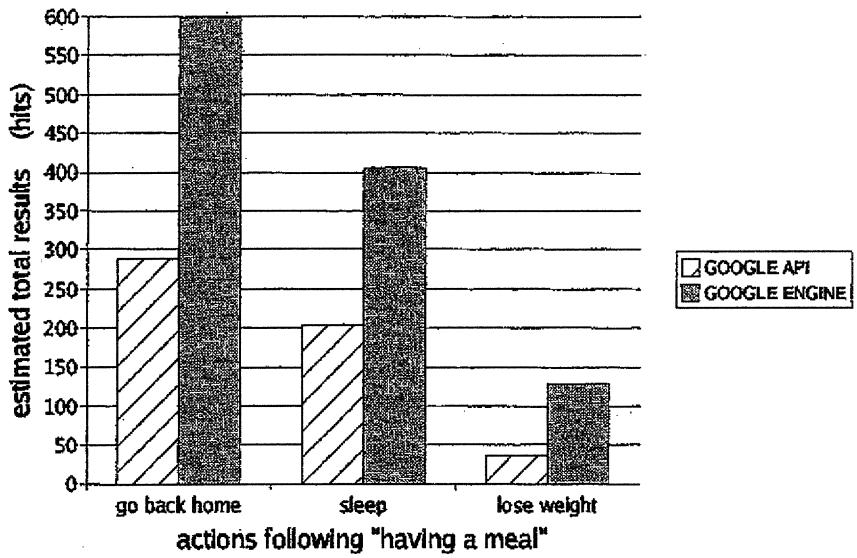


Fig. 1. Example: comparison for the data set of Google API and Google search engine.

3 Comparison with Other Languages

As mentioned in the introduction, we were interested how much the different language results differ from the Japanese ones. For example, in Japanese, we discovered that 3 most frequent actions which follow *watching a movie* are:
— watching movie → crying: 226 times
— watching movie → thinking: 135 times
— watching movie → being touched: 112 times
Then we used the Google searching engine for manual frequency checks for

English, Polish and Baidu searching engine for Chinese and we noticed that in most cases the proportions differ from Japanese results. The percentage of the example is shown in Fig.2. We confirmed that the 4 fullest sets of

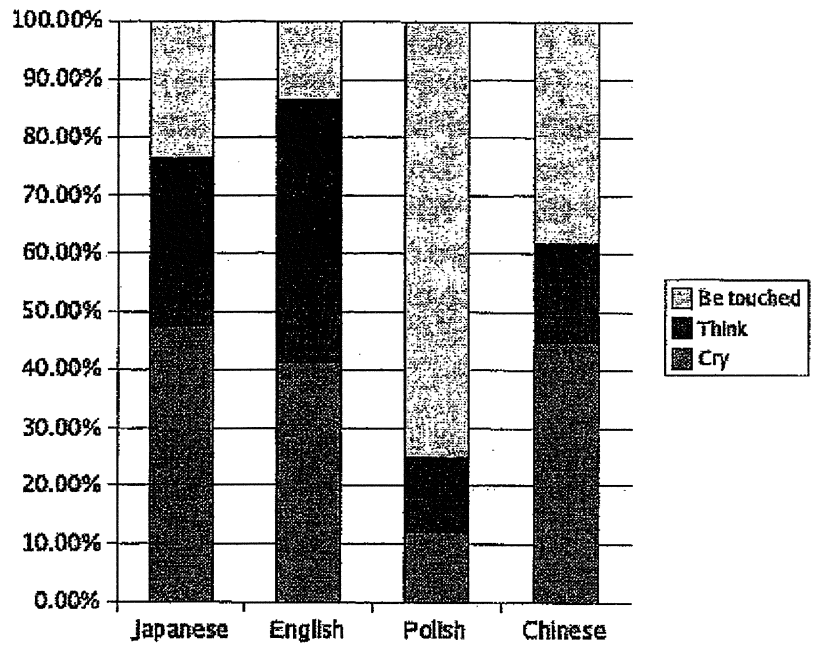


Fig. 2. Example: actions following "watching a movie" in Japanese, English, Polish and Chinese

trigrams had different proportions. Examined examples are shown in Tab. 1.

4 Possible Applications

Commonsensical knowledge data is very difficult to be collected manually as people in most cases do not realize things that happen too commonly. However, as our preliminary trials show, automatic retrieval of scripts might be useful to the researches in education, psycho-, socio-linguistics, economics (marketing), psychology and sociology. To suggest a few ideas:

- Language Education:
  — scripts can be used in automatic example sentences retrieval for foreign language students (they may differ depending on the context – if the categorization is successful);
  — when implemented into the talking agent, the script engine can correct the grammar of the learner and suggest the most common (natural) way of creating utterances;
- Economics and Marketing:
  — companies can retrieve differences of habits of their possible clients

Table 2. Proportions of actions in different languages

act1	verb2	JPN	ENG	POL	CHN
fill with water	boil	305	22	3	11900
	use	179	354	2	2
	mix	102	40	5	3710
take the subway	go home	120	14	7	311
	move	30	9	10	181
	went to	27	170	60	883
have a meal	go back home	288	6	0	769
	sleep	204	13	0	269
	lose weight	36	3780	48	1060
watch a movie	cry	226	1430	13	190
	think	135	1580	14	71
	be touched	112	467	81	162
drive a car	go back home	283	1590	517	7140
	move	180	125	17	67
	run away	64	276	35	117
hear a music	feel happy	150	424	1	318
	cry	119	752	99	659
	to be touched	82	1060	39	842
read a book	spend time	539	290	35	132
	study	392	588	41	34600
	think	240	2840	92	2590

depending on a culture;
— scripts made out of Internet resources are alive — the hit numbers can change together with customer's habits;

- Sociology, Anthropology:
— especially useful for comparative studies about developed countries and societies changing due to the Internet's growth.

5 Conclusions

We have created minimal scripts automatically by using Web resources and Web-mining techniques. We described our experiment with Japanese and we confirmed that Japanese language has comparatively useful structure for the commonsense retrieval from the enormous data sets and partially confirmed that the results would differ if we make the automatic script retrieval for other languages. This time we only compared Japanese top 3 scripts with the same scripts in 3 other languages but as we checked manually these top 3 results may vary depending on the language. We also shared our ideas how the results could be utilized by social sciences if the algorithms for other languages were developed.

6 Future Work

Currently we concentrate on creating noun categories made by verbs and combine simple scripts into the bigger, more complicated and more elastic strings which would be useful for creating plans and discovering goals. Our next step is to automatically retrieve causative relationships also for resolving problems which appear when a script is broken and something unusual happens. We suppose that also these results will differ depending on a culture and we plan to prove it.

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